SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING AND TECHNOLOGY, KATTANKULATHUR CAMPUS

DEPARTEMNT FO MECHANICAL ENGINEERING

Course Code	21EIC504J			**Course Website:**	
Course Title	REAL TIME EMBEDDED SYSTEMS			**Instructor:**	Dr. A. Xavier Reni Prasad
Semester	EVEN 2025-2026			**Email:**	xavierra@srmist.edu.in
Pre-requisite / Co-Requisite	NIL			**Slot & Class Hours:**	Slot A Day Order 1 & Day Order 3
Lecture Hall #	ESB409 - ELEC.SCIENCE	**Laboratory Location**	Control System Lab		

Organisation of the course

Classes will be predominatntly lectures with technology advancements and discussion on R&D works. Guest Lecturers from R&D are invited

Course Objectives and Learning Outcome(s)

1	Impart the knowledge on the basic concepts of embedded processor
2	Learn the programming techniques in ARM processor
3	Know the steps involved in designing an embedded system
4	Introduce the real time scheduling algorithms

Learning Materials

1. Marilyn Wolf,, High Performance embedded Computing: Applications in Cyber Physical Systems and Mobile Computing, 2 nd Edition, Elsevier 2014
2. JosephYiu,, The definitive Guide to ARM Cortex M3 and M4 Processors, 3 rd Edition, Elsevier 2020
3. Ata Elahi-Trever Arjeski, —ARM Assembly language with hardware experimentǁ, Springer Int. Publishing, 2016.
4. William hohl and Christoper Hinds, —ARM assembly language fundamentals and Techniques ǁCRC, 2 nd edition,2015.
5. Marilyn Wolf, Computers as Components: Principles of Embedded Computing System Design, Third Edition, Elsevier 2022
6. Daniel Kusswurm, Modern Arm Assembly Language Programming, Apress, 2020

Course Topics and Schedule

Week	Major Topic	Module / Unit	Method of delivery (Multi-Select)	Take home Assignment(s)	Learning materials to be referred
1	Review of Embedded Computing; Interfacing sensors embedded system design process; CPS and embedded Computing Architecture of ARM Cortex M3 and Cortex A series processors;	1	Lecture/Presentation		L1
2	ARM tool-chain for compilation, linking and execution phases; Memory system mechanism;	1	Lecture/Video		L4
3	Cache; Memory management units and address translation; Performance assessment of embedded processor; Introduction to Embedded Multicore Architecture;	1	Lecture/Presentation		L1
4	ARM virtual hardware: capabilities and use; QEMU for ARM processors	1	Individual tasks		L4
5	Interfacing sensors, Interfacing LED, Interfacing DAC, ADC Interfacing DIO, PWMs	1	Group activity	Assignment # 1	

6	Introduction set, Data transfer, Data processing, conditional and branch instructions, barrier and saturation operations,	2	Lecture/Presentation		L2
7	CortexM4-specific instructions, Thumb2 instructions, Programming of Embedded processors using assembly and C;	2	ped mode Discussion / Debate		L2
8	models for program -- data flow graphs; Assembly language programming of ARM Cortex M3; Hardware software co-design	2	ped mode Discussion / Debate		L2
9	Interfacing keyboard, Interfacing stepper motor, Interfacing Temperature sensor, Interfacing potentiometer sensor, Interfacing electrical switches	2	acticum / in-class assignm	Assignment # 2	L1,L2,L3
10	Design methodologies, Design flows, Requirement analysis, Multiple tasks and multiple processes,	3	Lecture/Presentation		L5
11	Preempt real time operating systems, Priority based scheduling,	3	Lecture/Video		L2
12	Distributed embedded systems, Examples of distributed embedded systems in industries, Wireless based embedded systems	3	Lecture/Presentation		L3
13	Interfacing process (time-driven), Interfacing event schedule, Debugging: Learn to find faults in a system, Tracing: Understand instruction set logs, derive problem loops	3	Practicum / in-class assignment	Assignment # 3	L5,L3
14	Processes and real time operating systems; Multi-rate system; real time scheduling algorithms - RMA,EDF and their variants;	4	Flipped mode Discussion / Debate		L5
15	Energy efficient scheduling algorithms; Data compression techniques, Examples of design of embedded systems. Architecture security features,	4	Individual tasks		L5
16	Arm Confidential Compute Architecture (Arm CCA) – an isolation technology that builds on the strong security foundations of TrustZone-AUTOSAR ECUs with ARM processors : Next generation HPCs and Zonal ECUs	4	Lecture/Presentation		L4
17	Communication between MSPs, Networking MSPs using Wi-Fi, Smart automation	4	Practicum / in-class assignment	Assignment # 4	L4,L5

…criteria

…ype of …ssessments	Periodicity	Weightage	Due Date (if applicable)	Remarks	
…J1-Test-10 Marks FJ2-Test-20 Marks	After 4th and 11th week respectively	30%	3-3-2026 to 12-3-2026; 28-4-2026 to 8-5-2026	Written Exam	
FJ1-Assignment 1&2 -5 Marks FJ2-Assignment 3&4 -10 Marks	Monthly once	15%	3-3-2026 to 12-3-2026; 28-4-2026 to 8-5-2026	Sum of all	
Practicum (Avg. of Experiments – 5 Marks, Practical exam & Viva – 10 Marks)	After 11th week	15%	28-4-2026 to 8-5-2026	Sum of all	
Continuous Assessment	**Total**	**60%**			
End Semester Examination		**40%**			
	Total	**100%**			

Make-up Policy

Assignments / Term Project

Proper academic performance depends on students doing their work not only well, but on time. Accordingly, assignments for this course must be received on the due date specified for the assignment. No Make-up will be available for assignments or term project. Late submissions will be evaluated at 25% less weight for that component for a delay of up to 24 hours after which no submissions will be accepted.

Test

Prior Permission of the course coordinator is usually required to get a make-up test.
A make-up test shall be granted only in genuine cases where, in the course faculty's judgment, the student would be physically unable to appear for the test. Course faculty's decision in this matter would be final.

Academic Integrity

Plagiarism or cheating will result in a grade of zero for the assignment or exam. Severe cases may result in course failure.

Additional Information

Provide a brief description (e.g. field trips, special lab session, special tutorials, industry certifications etc), dates, times, required materials or preparation, any fees or costs, etc.

Additional academic support

If any student requires additional free academic support (additional coaching), he/she may reach out to the Faculty Advisor

Notices

All notices concerning this course will be displayed online only. If there is a need srmist email would be used.

Contact Information

Feel free to reach out to Dr A. Xavier Reni Prasad via email for any questions regarding the course material. For urgent matters, you can visit the faculty room during office hours

Course Faculty

HOD